AI.212

Robot Kinematics

COURSE OVERVIEW

Describing robot motion mathematically: forward and inverse kinematics, Jacobians, and manipulator control.

TACT HOURS

dule

1

SESSION

60 MINUTES

ORIENTATION: MOTION IN MATH

KEY TOPICS

- Why kinematics matters
- Robot links and joints
- Degrees of freedom
- Workspace

TEACHING FOCUS

Describe robot geometry precisely.

KEY TAKEAWAY Kinematics is the language of robot motion.

ROBOT KINEMATICS

2

SESSION

60 MINUTES

REPRESENTATIONS

KEY TOPICS

- Rotation matrices
- Homogeneous transforms
- Euler angles and quaternions
- Denavit–Hartenberg convention

TEACHING FOCUS

Represent position and orientation.

KEY TAKEAWAY Transforms move frames through space.

ROBOT KINEMATICS

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SESSION

60 MINUTES

FORWARD KINEMATICS

KEY TOPICS

- Computing end-effector pose
- DH parameters
- Serial-chain FK
- Examples: 2-link, 6-DOF arm

TEACHING FOCUS

Find where the tool is from joint angles.

▲ **KEY TAKEAWAY** Forward kinematics answers 'where am I?'.

4

SESSION

60 MINUTES

INVERSE KINEMATICS

KEY TOPICS

- IK problem
- Analytical vs. numerical IK
- Multiple solutions
- Jacobian-based IK

TEACHING FOCUS

Find joint angles for a desired pose.

KEY TAKEAWAY Inverse kinematics answers 'how do I get there?'.

5

SESSION

60 MINUTES

JACOBIANS & VELOCITIES

KEY TOPICS

- Jacobian matrix
- Velocity kinematics
- Singularities
- Manipulability

TEACHING FOCUS

Relate joint speeds to tool speed.

KEY TAKEAWAY The Jacobian maps motion spaces.

6

SESSION

60 MINUTES

KINEMATICS IN CONTROL & REVIEW

KEY TOPICS

- Kinematic control of manipulators
- Course recap and assessment
- Toward mobile robotics (AI.207)
- IK lab

TEACHING FOCUS

Use kinematics to command robot arms.

KEY TAKEAWAY Control without kinematics is blind.